

Method	Reference / oracle	Independence	What it catches
Fixed KATs	X-Wing draft vectors; RFC 7748	published vectors	combiner & X25519 spec-conformance regressions
NIST ACVP	<i>usnistgov</i> ACVP, full FIPS family	NIST ground truth	primitive non-conformance (FIPS 203/204/205)
Multi-backend differential	RustCrypto <code>ml-kem/ml-dsa</code> ; orion X25519	independent re-impls.	implementation bugs (must stay byte-identical)
EasyCrypt proof	machine-checked; CR(SHA3)	formal (computational)	binding-design flaws in the combiner
Cross-platform realization	deterministic vectors + product invariants	evidence mode per cell	byte drift in conformance; policy/context/state/failure drift in ABI 2 product cells
Generative property tests	proptest: 9 props $\times$ 128 cases	randomized	edge-case violations (injectivity, dom. sep., K-CTX)